F&B AI HQ

A 1-Man F&B Headquarters Dashboard Framework

Printable strategy map for building an AI-powered operating control room across finance, HR, procurement, POS, and marketing.

Core idea

The product is not merely an AI dashboard. It is a daily profit control system that helps an F&B owner see where money is leaking and what action to take immediately.

The system should answer four questions every day:

1. Did we make money yesterday?
2. Where did we leak money?
3. Which outlet, SKU, staff shift, supplier, or campaign caused it?
4. What should the owner do today?

1. Executive Summary

Best wedge
Start with the Daily Prime Cost Dashboard: sales, food cost, labour cost, prime cost, and recommended actions. This is more valuable than a full accounting dashboard at the beginning.

The strongest positioning is:

- AI-powered F&B profit control system for owner-operated restaurants.
- A 1-man HQ for F&B owners who need daily visibility on food cost, labour cost, and profit leakage.
- A daily F&B profit control room, not another reporting dashboard.

The system should connect existing F&B tools rather than replacing them at the start. It should pull from POS, payroll/timesheets, supplier invoices, recipe costing, inventory, accounting, and marketing spend. The AI layer should interpret, flag issues, and recommend action.

2. Core Architecture

Layer	Function	Examples
Data Sources	Raw business inputs	POS sales, supplier invoices, staff timesheets, recipes, bank transactions, marketing spend
Processing Layer	Clean, match, and calculate	Item matching, recipe costing, wage calculation, GST categorisation, inventory movement
Agent Layer	Interpret and recommend	Sales analysis, food cost variance, roster optimisation, cashflow warnings
Dashboard	Show decisions clearly	Daily owner snapshot, department dashboards, red flags
Action System	Draft or execute controlled actions	Generate PO, draft supplier order, draft roster, send weekly report

Important warning
Do not start by finding agents. Start by defining the business control system, master data, and workflows. Agents are only useful when the data model is clean.

3. The 5 HQ Departments

Department	Purpose	Primary Question
Accounts / Finance	Show whether the business is profitable and solvent.	Did we make money and can we pay what is due?
HR / Manpower	Control labour cost by hour, daypart, role, and outlet.	Were we overstaffed or understaffed for the sales volume?
Procurement / Inventory	Control purchase prices, wastage, stock, and COGS.	Did buying, usage, or wastage damage margin?

POS / Sales / Menu Engineering	Understand what sells, when it sells, and whether it is profitable.	Which items and hours drive profit?
Marketing / Growth	Link campaign spend to revenue and customer behaviour.	Did marketing create incremental sales or just discount existing customers?

4. Accounts / Finance

Purpose: show whether the business is actually profitable and whether cashflow is safe.

Function	What It Tracks	Output
AP / Accounts Payable	Supplier bills, rent, utilities, subscriptions, contractors	What the business owes
AR / Sales Receivables	POS takings, delivery payouts, event invoices	What the business is owed
Cashflow	Bank balance, upcoming payments, expected receipts	Survival runway
P&L	Revenue, COGS, labour, rent, utilities, marketing, admin	Profit and loss
Balance Sheet	Assets, liabilities, loans, inventory value	Financial health
Tax / GST	GST collected, GST paid, GST payable	Compliance
Management Reports	Daily, weekly, monthly performance	Owner visibility

- Key metrics: daily sales, gross profit, net profit, COGS %, labour %, rent %, prime cost %, cash balance, payables due, receivables due, GST payable.
- Most important F&B metric: Prime Cost = Food Cost + Labour Cost. If prime cost is unhealthy, the business is structurally weak.

Agent	Job
Invoice Reader Agent	Reads supplier invoices and extracts amount, GST, item, supplier, and due date.
Expense Categorisation Agent	Classifies costs into food, beverage, labour, rent, utilities, marketing, and admin.
P&L Agent	Generates daily, weekly, and monthly profit reports.
Cashflow Agent	Predicts upcoming cash crunches.
Variance Agent	Flags unusual cost spikes.
GST Agent	Tracks GST collected versus claimable GST.
Owner Summary Agent	Turns finance results into plain-English daily summaries.

5. HR / Manpower

Purpose: control labour cost by daypart, outlet, role, and sales volume.

Function	What It Tracks	Output
Full-timer salaries	Basic pay, CPF, benefits, leave	True fixed labour cost
Part-timer timesheets	Clock-in/out, hourly rate, outlet, role	Actual daily labour cost
CPF	Employer and employee CPF	Statutory cost
FWL	Work permit / S Pass levy	True foreign manpower cost
Leave / MC / OT	Absence, overtime, replacement cost	Scheduling pressure
Roster	Planned manpower by hour	Planned labour cost
Attendance	Actual worked hours	Actual labour cost
Productivity	Sales per labour hour, covers per labour hour	Staff efficiency

- Key metrics: labour cost %, sales per labour hour, covers per labour hour, revenue per staff, part-time cost per day, full-time daily cost allocation, OT cost, manpower cost by hour.
- A powerful metric is Manpower Cost Per Trading Hour, e.g. 11am-12pm sales of $280 against staff cost of $96 gives labour cost of 34.3%.

Agent	Job
Timesheet Agent	Pulls clock-in/out data and calculates wages.
CPF/FWL Agent	Estimates statutory labour cost.
Roster Optimisation Agent	Compares roster versus actual sales patterns.
Labour Variance Agent	Flags overstaffed or understaffed periods.
Productivity Agent	Calculates sales per labour hour.
Compliance Agent	Tracks leave, rest days, overtime, and MOM-related issues.
Staffing Recommendation Agent	Suggests roster changes.

6. Procurement / Inventory

Purpose: control food cost, supplier pricing, wastage, and stock movement.

Function	What It Tracks	Output
Supplier database	Supplier, terms, contact, category	Vendor control
Item master	SKU, unit, pack size, price, GST, supplier	Purchase control
Price list	Historical buying price	Price movement
Purchase Order	Items selected to order	PO generation
Delivery Order	Items received	Stock-in
Invoice Matching	PO versus DO versus invoice	Price and quantity discrepancy
Inventory	Opening stock, purchases, usage, closing stock	Stock value
Wastage	Spoilage, staff meal, mistakes	True COGS
Stock alerts	Low stock, overstock, dead stock	Reorder control

- Key metrics: daily purchases, inventory value, stock variance, actual versus theoretical usage, supplier price increase, wastage %, dead stock, reorder level, COGS %.
- Theoretical COGS = based on recipes and sales. Actual COGS = opening stock + purchases - closing stock. Variance indicates leakage, wastage, theft, portioning issues, or recipe errors.

Agent	Job
Supplier Price Agent	Tracks item price increases.
PO Generator Agent	Creates suggested purchase orders.
Invoice Matching Agent	Checks PO versus DO versus invoice.
Inventory Agent	Updates stock levels.
Wastage Agent	Tracks spoilage and abnormal usage.
Reorder Agent	Suggests what to buy based on par levels.
Supplier Negotiation Agent	Flags items to renegotiate.
COGS Variance Agent	Compares theoretical versus actual food cost.

7. POS / Sales / Menu Engineering

Purpose: understand what sells, when it sells, and whether each item is profitable.

Function	What It Tracks	Output
Menu item master	SKU, selling price, category, tax, modifiers	Sales control
Sales by item	Quantity sold, revenue, discounts	Product performance
Sales by hour	Daypart revenue	Staffing insight
Sales by channel	Dine-in, takeaway, delivery, events	Channel profitability
Recipe costing	Ingredient cost per item	Gross margin
Modifier costing	Add-ons and substitutions	True item profitability
Discounts / voids	Promos, staff discounts, mistakes	Revenue leakage
Menu engineering	Popularity versus margin	What to push or remove

- Key metrics: sales per hour, sales per daypart, average transaction value, average spend per pax, item quantity sold, gross margin per item, food cost per item, contribution margin, discount %, void %, channel sales mix.
- The key menu metric is not only food cost %. Contribution Margin = Selling Price - Food Cost. A high food-cost item may still contribute more dollars than a low food-cost item.

Menu Category	Meaning	Action
Star	High sales, high margin	Push harder
Plowhorse	High sales, low margin	Reprice or reduce cost
Puzzle	Low sales, high margin	Market better, rename, or reposition
Dog	Low sales, low margin	Remove or redesign

Agent	Job

Sales Import Agent	Pulls POS sales data.
Sales Pattern Agent	Finds peak and slow hours.
Menu Costing Agent	Connects recipe cost to selling price.
Menu Engineering Agent	Classifies stars, puzzles, plowhorses, and dogs.
Discount Leakage Agent	Flags excessive discounts and voids.
Channel Profitability Agent	Compares dine-in versus delivery margin.
Sales Forecast Agent	Predicts demand by day, hour, weather, and event.
Pricing Agent	Suggests price changes.

8. Marketing / Growth

Purpose: connect marketing spend to actual revenue and customer behaviour.

Function	What It Tracks	Output
Campaign spend	Meta, Google, TikTok, influencer, PR	Marketing cost
Promo codes	Campaign attribution	Sales linked to campaign
Reservation source	Google, Instagram, Chope, walk-in	Channel source
Customer database	Repeat visits, spend, birthdays	CRM
Loyalty	Points, stamps, rewards	Retention
Reviews	Google reviews, delivery ratings	Reputation
Campaign ROI	Spend versus sales lift	Marketing efficiency

- Key metrics: marketing spend, sales uplift, customer acquisition cost, return on ad spend, repeat customer rate, reservation source, promo redemption, review rating, cost per booking, cost per new customer.
- F&B attribution is messy. Use direct attribution plus estimated sales lift rather than pretending every sale can be perfectly attributed.

Agent	Job
Campaign Tracking Agent	Tracks marketing spend by platform.
Promo Code Agent	Links promos to sales.
Review Monitoring Agent	Summarises Google and delivery reviews.
CRM Agent	Tracks repeat customers.
Content Agent	Suggests campaigns based on slow-moving or high-margin items.
ROI Agent	Compares spend to revenue lift.
Local Demand Agent	Flags events, weather, holidays, and nearby demand shifts.

9. Master Data Model

This is the foundation. If master data is weak, AI conclusions will be unreliable.

Master Table	Fields Needed

Outlet master	Outlet name, address, opening hours, seating capacity, rent, service charge, operating days, cost targets
Supplier master	Supplier name, category, payment terms, contact person, GST status, delivery days, minimum order
Item master	Item name, supplier, category, purchase unit, pack size, unit cost, GST, storage type, shelf life, reorder level
Recipe master	Menu item, ingredient, quantity used, unit cost, yield %, wastage %, recipe cost, selling price, food cost %, contribution margin
Staff master	Name, role, employment type, hourly/monthly rate, CPF status, FWL status, outlet, start date, leave entitlement
Sales master	Date, time, outlet, receipt number, item sold, quantity, selling price, discount, tax, payment method, channel
Marketing master	Campaign, platform, spend, start date, end date, promo code, target item, target audience, revenue attributed

10. Combined Metrics: Where the Power Is

Combination	Calculation / Insight
Finance + POS	Revenue by daypart, item, channel, and target variance flows into P&L.
POS + Procurement	Items sold translate into ingredients consumed, theoretical COGS, inventory depletion, and reorder needs.
HR + POS	Sales by hour are compared against labour by hour to identify overstaffing and understaffing.
Marketing + POS	Marketing spend is compared against promo-linked sales and estimated sales uplift.
Procurement + Finance	Invoices flow into AP, actual COGS, supplier inflation, and cashflow pressure.

11. Operating Control Reports

Level	Report	Purpose
Level 1	Owner Daily Snapshot	One-page view of sales, food cost, labour cost, prime cost, cash, red flags, and actions.
Level 2	Department Dashboards	Finance, HR, procurement, POS, and marketing drill-down.
Level 3	Detailed Reports	Sales by hour, labour by hour, sales by item, channel profitability, supplier trends, inventory variance, marketing ROI.

Daily owner snapshot example
Sales: $5,420. Food cost: 33.8%. Labour cost: 29.6%. Prime cost: 63.4%. Red flags: seafood cost up, 3pm-5pm overstaffed, pasta margin dropped. Actions: reduce Tuesday afternoon PT shift, raise vongole price by $1, reorder chicken from Supplier B.

12. Agent Architecture

Agent Group	Agents
Data Intake Agents	POS Import, Invoice Reader, Timesheet Import, Bank Feed, Inventory Count, Marketing Spend Import
Processing Agents	Data Cleaning, Item Matching, Recipe Costing, Payroll Cost, GST/Tax, Inventory Movement, P&L Builder
Analysis Agents	Sales Analysis, Food Cost, Labour Cost, Prime Cost, Menu Engineering, Procurement Variance, Cashflow, Marketing ROI
Recommendation Agents	Pricing Recommendation, Roster Recommendation, Purchase Recommendation, Menu Optimisation, Supplier Negotiation, Cost Reduction
Action Agents	Generate PO, Send supplier order, Draft payroll summary, Create weekly report, Alert owner, Draft staff roster, Draft price-change proposal

Control principle
In the early product, agents should recommend and draft. They should not auto-execute supplier orders, payroll, payments, or price changes without owner approval.

13. MVP: What to Build First

Do not build accounting, HR, procurement, POS, and marketing all at once. Build one painful use case first: the Daily Prime Cost Dashboard.

Input	Output
POS sales	Daily sales, sales by hour, top-selling items, channel mix
Recipe costing	Estimated food cost, item margin, low-margin items
Staff timesheets	Daily labour cost, labour by hour, overstaffed periods
Supplier invoices	Supplier price changes and purchase cost alerts

- MVP dashboard: sales yesterday, labour cost yesterday, estimated food cost yesterday, prime cost yesterday, top 5 best sellers, top 5 worst-margin items, staff cost by hour, food cost red flags, recommended actions.
- First six agents: POS Import Agent, Recipe Costing Agent, Timesheet Labour Agent, Supplier Invoice Agent, Daily P&L / Prime Cost Agent, Owner Recommendation Agent.

14. Phase Roadmap

Phase	Goal	Build
Phase 1: Visibility	Show what happened.	POS import, sales report, timesheet cost, recipe cost, daily prime cost, basic dashboard
Phase 2: Control	Identify waste and leakage.	Invoice upload, supplier price tracking, inventory tracking, theoretical vs actual COGS, labour by hour, menu margin analysis
Phase 3: Recommendations	Tell owner what to do.	Roster recommendations, price change suggestions, supplier switch suggestions, menu engineering, order quantity suggestions
Phase 4: Action	Execute controlled workflows.	Generate PO, send supplier order, draft roster, weekly owner report, accountant handover, marketing campaign draft
Phase 5: Benchmarking	Compare outlets and businesses.	Outlet vs outlet, cuisine benchmark, labour benchmark, food cost benchmark, sales per sqm, sales per seat, revenue per labour hour

15. Correct Build Sequence

5. Build a spreadsheet prototype.
6. Create master data: menu, recipes, suppliers, ingredients, staff.
7. Import POS sales from CSV before API integration.
8. Calculate theoretical food cost.
9. Import timesheets.
10. Calculate labour cost by hour.
11. Generate daily prime cost report.
12. Add supplier invoice reading.
13. Add price variance alerts.
14. Add AI daily summary.
15. Add PO generation.
16. Add inventory tracking.
17. Add marketing ROI.
18. Add accounting integration.

Why accounting comes later
Accounting data is usually monthly. F&B needs daily operating control. The system should not wait for the accountant to close the books before the owner knows whether the business is bleeding.

16. First Prototype Table Structure

Table	Purpose
Outlets	Location and cost targets
Suppliers	Vendor terms and categories
Ingredients	Item master and purchase cost
Menu Items	Selling price, category, channel, tax
Recipes	Ingredient quantities, yield, wastage, food cost
POS Sales	Sales transactions by item, time, channel
Purchases	Supplier invoices, PO, DO, price
Inventory	Opening, purchase, usage, closing stock
Staff	Role, employment type, cost rate, statutory status
Timesheets	Clock-in/out and actual labour hours
Marketing Spend	Campaign spend and promo attribution
Daily Metrics	Sales, food cost, labour cost, prime cost
AI Recommendations	Red flags and actions

17. Practical Tech Stack

Stage	Recommended Tools
Prototype	Google Sheets or Airtable, Looker Studio or Retool, Make/Zapier/n8n, OpenAI or Claude, OCR invoice reader, POS CSV imports
Proper Product	React or Retool frontend, Supabase/PostgreSQL database, n8n/Make automations, OpenAI/Claude AI layer, Google Document AI/Mindee/Azure Form Recognizer for OCR
Integrations Later	Xero/QuickBooks, Talenox/Payboy/QuickHR, Qashier/Lightspeed/Shopify POS/Oddle/delivery exports

Recommended first stack

Airtable + Make/n8n + OpenAI + Retool/Looker Studio. It is fast, flexible, relatively cheap, and sufficient to prove demand before building custom software.

18. Critical Warnings

Warning	Meaning
POS data alone is not enough	POS shows what sold, not whether the business made money. You need recipe costing and labour data.
Accounting data is too slow	Monthly accounts do not help an owner act today.
Live inventory is hard	Wastage, staff meals, substitution, yield loss, spoilage, theft, and inconsistent portioning make live inventory messy. Start with theoretical inventory first.

AI cannot fix bad master data	If item names, supplier invoices, recipes, or pack sizes are messy, AI will produce wrong conclusions.
Owners do not want more dashboards	They want decisions. The system must recommend actions, not just show charts.

19. Example AI Owner Briefs

Daily brief

Yesterday sales were $5,420, 8% below last Tuesday. Labour cost was 34%, above the 28% target. Main issue: 3pm-5pm had low sales but 4 staff on shift. Food cost was estimated at 31.5%, slightly above target. Recommended actions: reduce 1 PT shift next Tuesday afternoon, review salmon bowl pricing, shift marketing push to high-margin pasta items.

Weekly brief

This week sales increased 12%, but profit did not improve because labour rose 18% and seafood cost rose 9%. Best-performing item: chicken pasta, high sales and 24% food cost. Worst-performing item: burrata salad, low sales, 42% food cost, high wastage. Recommended actions: remove burrata salad or make it weekend-only, increase vongole by $1.50, reduce weekday lunch roster by 6 PT hours.

20. Business Positioning

- Do not position it as generic SaaS at the start. Position it as an AI-powered F&B profit control system.
- The strongest pain: most small F&B owners only know if they made money after the month-end accounts. This system shows daily leakage.
- Charge for implementation, not just software. F&B data is messy, and owners need help cleaning recipes, suppliers, payroll assumptions, and dashboards.
- Your advantage is F&B operating judgment. The AI can calculate, but your credibility explains what the numbers mean and what the owner should do.

21. Immediate Next-Step Checklist

19. Choose one pilot restaurant or mock concept.
20. Create the item master, supplier master, recipe master, staff master, and sales import template.
21. Import one week of POS sales.
22. Cost the top 20 menu items by sales volume.
23. Import one week of timesheets.
24. Calculate daily food cost, labour cost, and prime cost.
25. Generate a one-page daily owner snapshot.
26. Add an AI summary that produces red flags and actions.
27. Test whether the owner would pay for this insight.
28. Only then build integrations and automations.